NPC – UNIT 4 ASSIGNMENT

Configure DHCP and Set Up a Network Using Cisco Packet Tracer

Student Name	Swapnil Shejwal
PRN	2125UCSM1043
Subject / Module	Network Protocols and Communication (NPC) - Unit 4
Project Title	DHCP Configuration and Network Setup
Software / Tools	Cisco Packet Tracer 9.0.1 (Windows/Kali)
GitHub Repository	Configure-DHCP-and-Set-Up-a-Network-Using-Cisco-Packet-Tracer
GitHub Link	https://github.com/SwapnilShinde148/Configure-DHCP-and-Set-Up-a-Network-Using-Cisco-Packet-Tracer

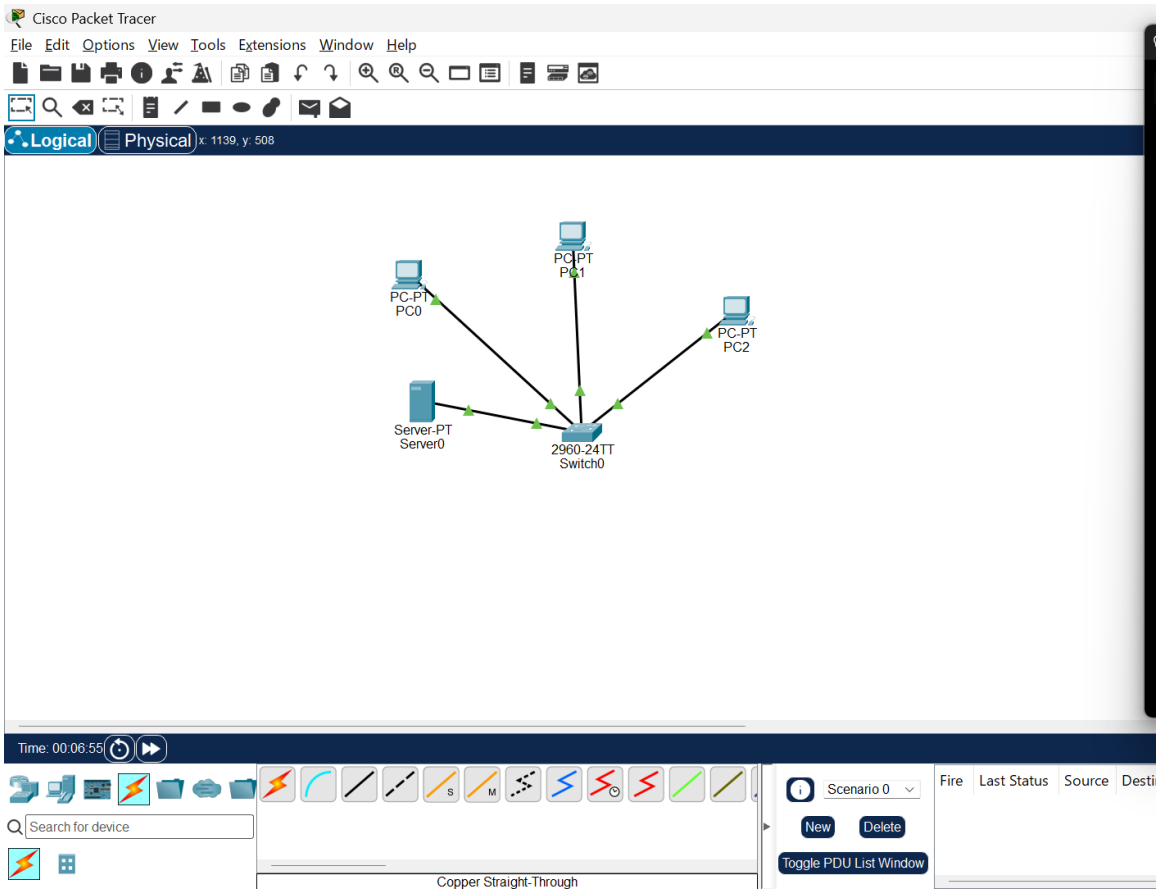
1. Aim

To create a small LAN in Cisco Packet Tracer, configure a server with a static IPv4 address, enable the DHCP service, assign IP configuration automatically to client PCs, and verify end-to-end network connectivity using the ICMP ping command.

2. Network Topology

The topology contains three PCs (PC0, PC1, PC2), one Cisco 2960-24TT Switch (Switch0), and one Generic Server (Server0). Each end device is connected to the switch using Copper Straight-Through Ethernet cables.

Figure 1: Network topology with PC0, PC1, PC2, Server0, and Switch0.

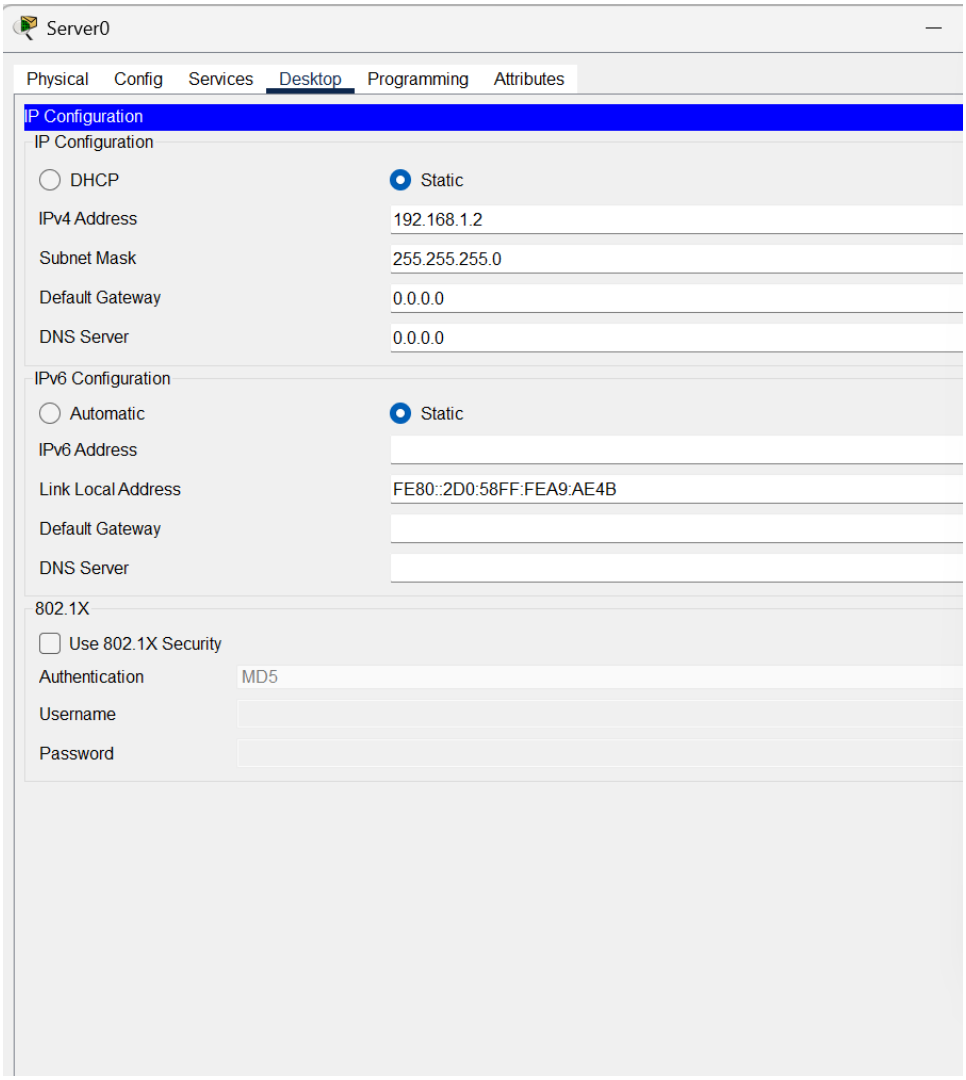


3. IP Addressing and DHCP Configuration

Server0 was configured with a static IPv4 address so that clients can communicate with it reliably. The DHCP service was enabled and a pool named NPC_POOL was configured.

Device / Setting	Configured Value
Server0 IPv4 address	192.168.1.2
Subnet mask	255.255.255.0
DHCP pool name	NPC_POOL
DHCP pool start address	192.168.1.10
DHCP pool subnet mask	255.255.255.0
Maximum number of users	50
Default gateway / DNS	0.0.0.0 (Local LAN)

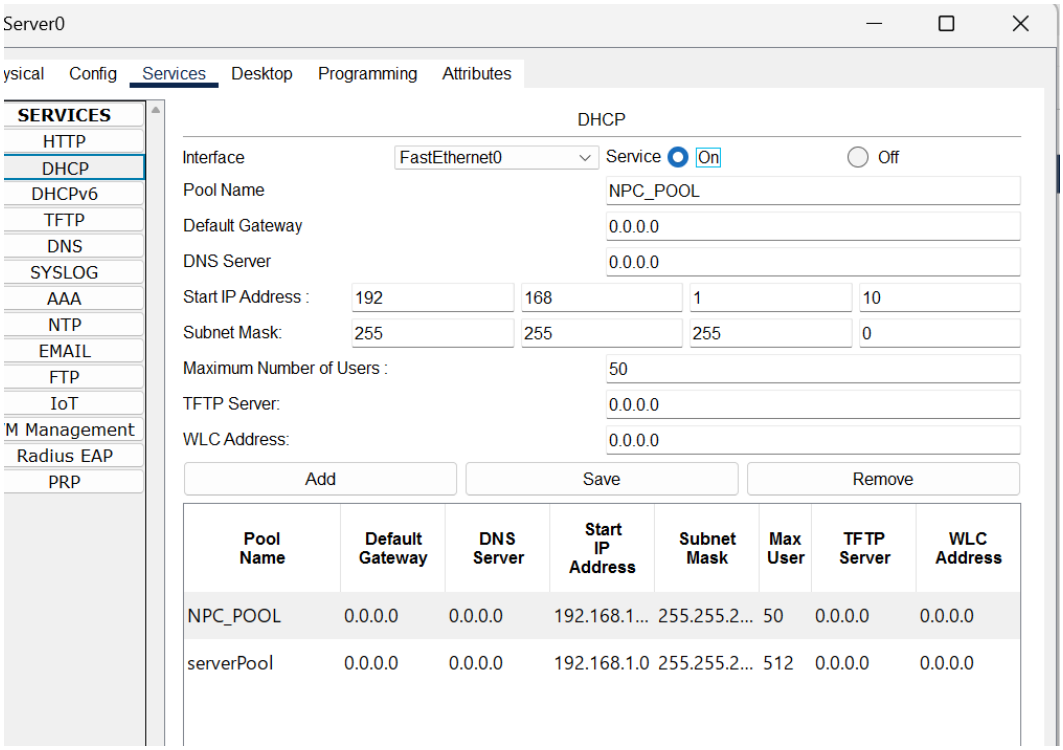
Figure 2: Server0 static IP configuration.



4. DHCP Service Setup

In Server0, Services > DHCP was opened. The DHCP service was switched On, and the NPC_POOL entry was configured with the start IP address 192.168.1.10 and subnet mask 255.255.255.0.

Figure 3: DHCP service and NPC_POOL settings.



5. Automatic IP Configuration on PCs

On each PC, Desktop > IP Configuration was opened and DHCP was selected. The screenshots show successful DHCP requests (DHCP request successful) and the dynamic IPv4 addresses received by the PCs.

Figure 4: PC0 IP configuration (DHCP successful; IPv4 address: 192.168.1.1).

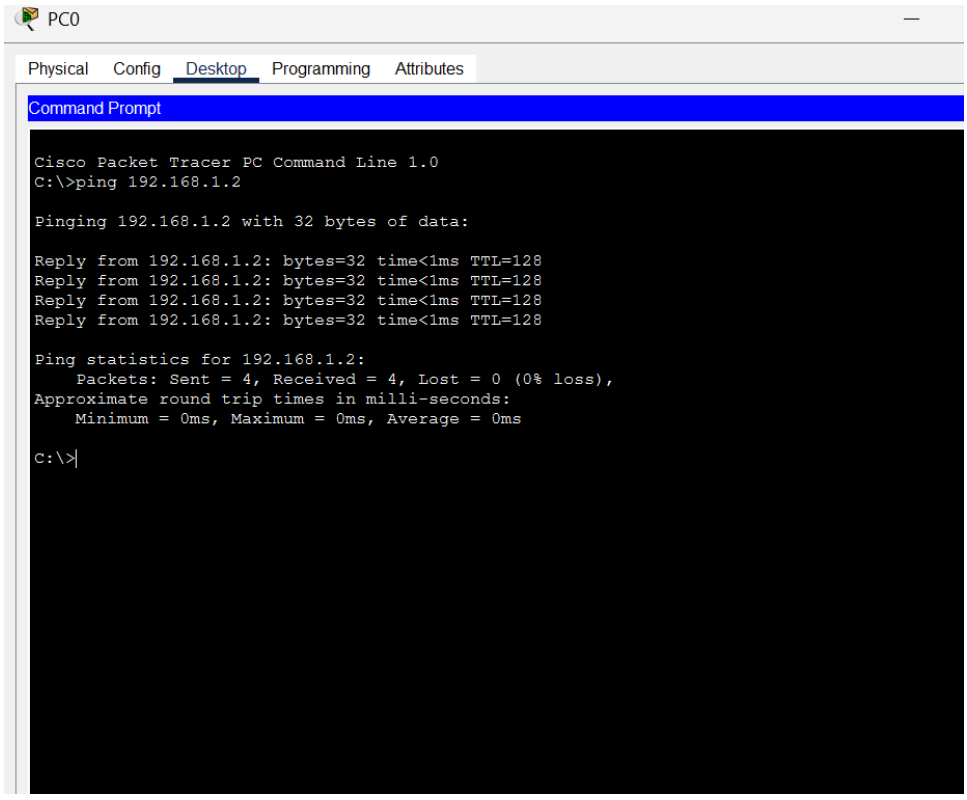


Figure 5: PC1 IP configuration (DHCP successful; IPv4 address: 192.168.1.3).

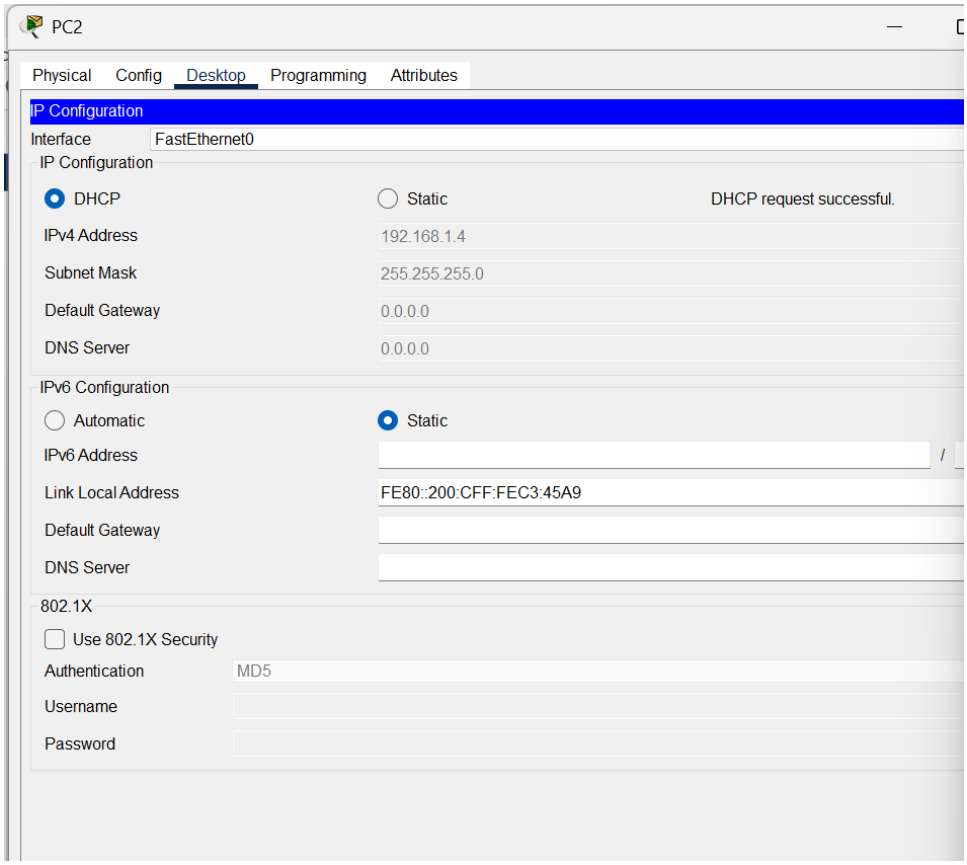
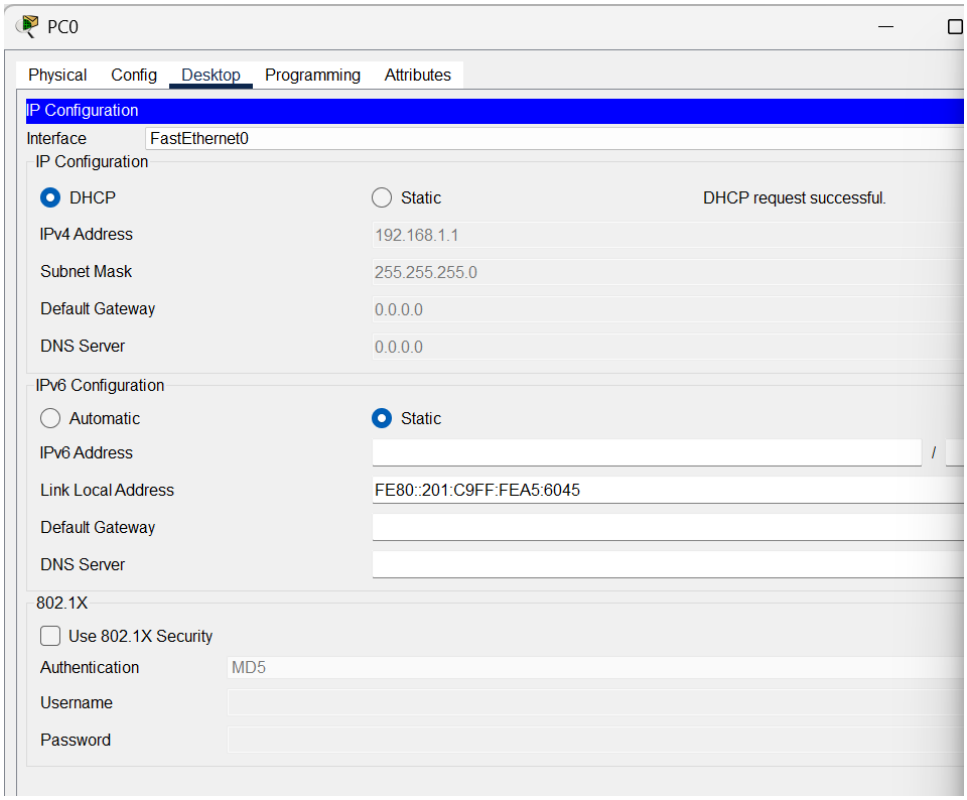


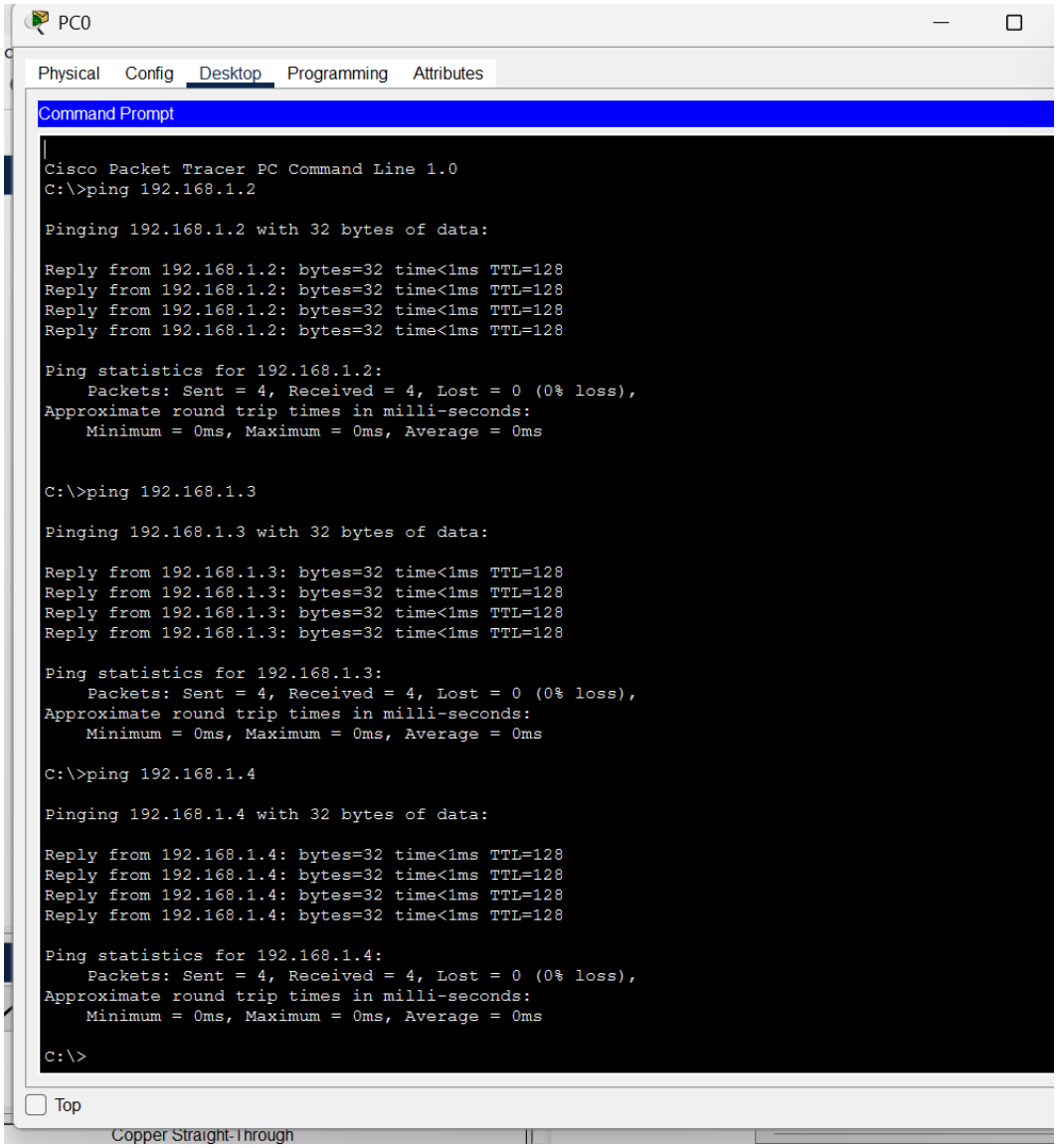
Figure 6: PC2 IP configuration (DHCP successful; IPv4 address: 192.168.1.4).



6. Connectivity Testing

Network connectivity was verified from PC0 using the ICMP ping command in the Command Prompt. The ping tests to Server0 (192.168.1.2), PC1 (192.168.1.3), and PC2 (192.168.1.4) all returned replies with 4 packets sent, 4 received, and 0% packet loss.

Figure 7: Successful ping tests from PC0 to Server0, PC1, and PC2.



7. Results and Conclusion

The LAN topology was created successfully in Cisco Packet Tracer. Server0 was assigned a static IP address, the DHCP service was enabled, and the client PCs received IPv4 configuration automatically. Ping tests confirmed flawless connectivity between PC0, Server0, PC1, and PC2 with 0% packet loss.

8. Submission Evidence Checklist

✓ Cisco Packet Tracer Project File: NPC_Unit4_DHCP_Assignment.pkt

✓ Project Documentation Report: NPC_Unit4_DHCP_Assignment_Report.pdf & .docx

✓ Complete Topology & Configuration Screenshots in screenshots/ folder

✓ GitHub Repository: https://github.com/SwapnilShinde148/Configure-DHCP-and-Set-Up-a-Network-Using-Cisco-Packet-Tracer